$\{\because \alpha v_1 + \beta v_2 \in R(T)$

$\alpha v_1 + \beta v_2 = \alpha T(u_1) + \beta T(u_2)$

$= T(\alpha u_1 + \beta u_2)$

$\therefore \alpha v_1 + \beta v_2 \in R(T)$

Q:-1 $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$T(x,y,z) = (0,y,z)$

Let $u \in \mathbb{R}^3$ st $T(u) = 0$, $u = (x,y,z)$

$T(x,y,z) = (0,0,0)$

$\Rightarrow (0,y,z) = (0,0,0) \Rightarrow y = z = 0$, x is arbitary

Let $x = \alpha$

$\therefore Ker(T) = \{(x,0,0) \in \mathbb{R}^3\}$

$= \{\alpha(1,0,0), \alpha \in \mathbb{R}\}$

$R(T) = \{(0,y,z) \in \mathbb{R}^3 \mid T(x,y,z) = (0,y,z)\}$

$=$ YZ-plane.

Q:-2 $T(x,y,z) = (x+y+z, x+y, x+y)$

Let $T(x,y,z) = (0,0,0)$

$\Rightarrow x+y+z = 0$

$x+y = 0$

$x+y = 0$

$\Rightarrow x+y+z = 0$

$x+y = 0$

$\Rightarrow x+y+z = 0$

$x = -y$

P Kalika Maths